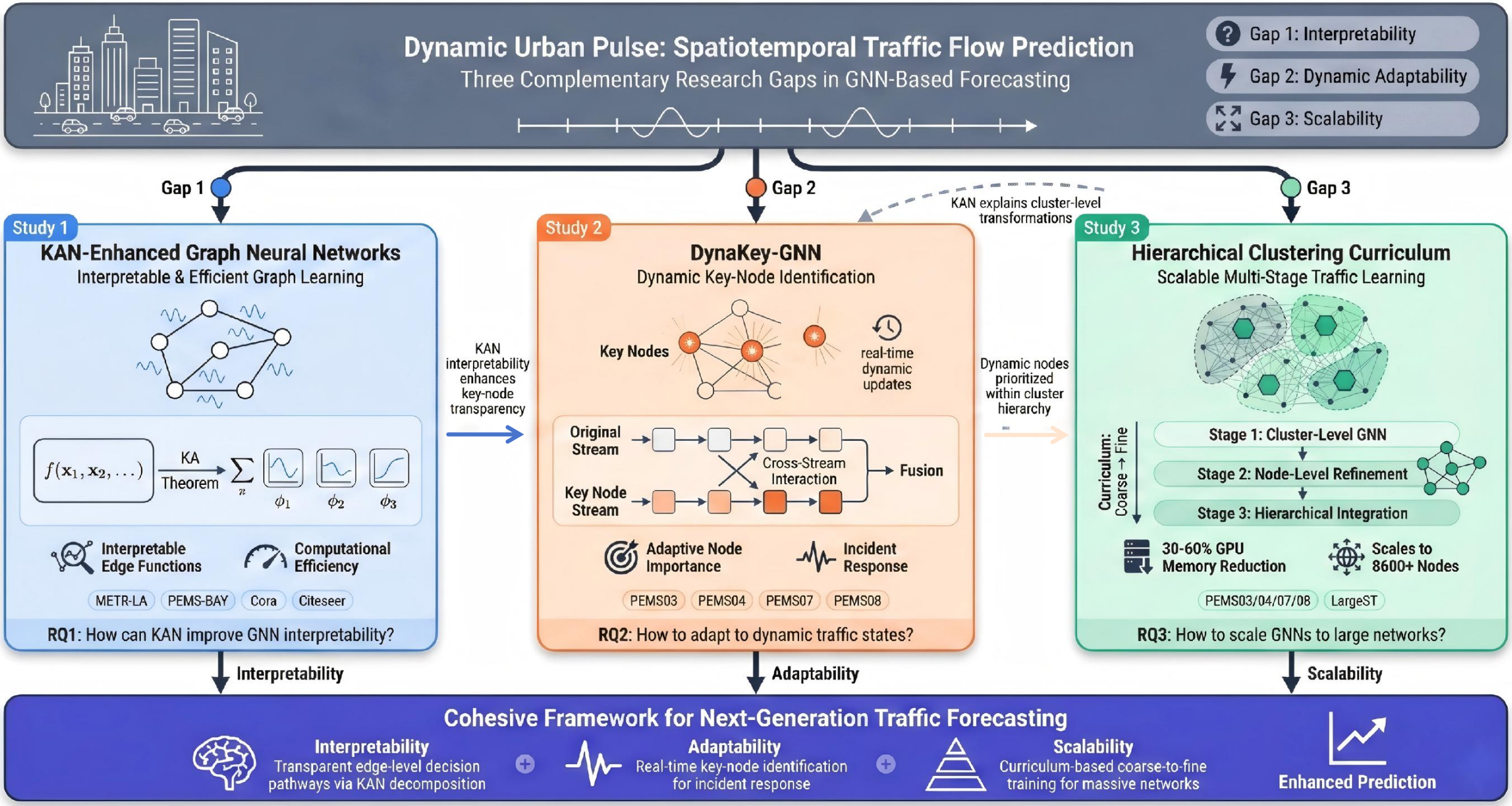
Dynamic Urban Pulse: Spatiotemporal Traffic Flow Prediction
Three Complementary Research Gaps in GNN-Based Forecasting
Gap 1: Interpretability
Gap 2: Dynamic Adaptability
Gap 3: Scalability
Gap 1
Gap 2
Gap 3
KAN explains cluster-level transformations
Study 1
KAN-Enhanced Graph Neural Networks
Interpretable & Efficient Graph Learning
$f(x_1, x_2, \ldots)$
KA Theorem
$\sum_n$
ϕ_1 ϕ_2 ϕ_3
Interpretable Edge Functions
Computational Efficiency
METR-LA
PEMS-BAY
Cora
Citeseer
RQ1: How can KAN improve GNN interpretability?
KAN interpretability enhances key-node transparency
Study 2
DynaKey-GNN
Dynamic Key-Node Identification
Key Nodes
real-time dynamic updates
Original Stream
Key Node Stream
Cross-Stream Interaction
Fusion
Adaptive Node Importance
Incident Response
PEMS03
PEMS04
PEMS07
PEMS08
RQ2: How to adapt to dynamic traffic states?
Dynamic nodes prioritized within cluster hierarchy
Study 3
Hierarchical Clustering Curriculum
Scalable Multi-Stage Traffic Learning
Curriculum: Coarse → Fine
Stage 1: Cluster-Level GNN
Stage 2: Node-Level Refinement
Stage 3: Hierarchical Integration
30-60% GPU Memory Reduction
Scales to 8600+ Nodes
PEMS03/04/07/08
LargeST
RQ3: How to scale GNNs to large networks?
Interpretability
Adaptability
Scalability
Cohesive Framework for Next-Generation Traffic Forecasting
Interpretability
Transparent edge-level decision pathways via KAN decomposition
Adaptability
Real-time key-node identification for incident response
Scalability
Curriculum-based coarse-to-fine training for massive networks
Enhanced Prediction